

Candidate Name	CT Group	Candidate Number



VICTORIA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2008

H2 CHEMISTRY

9746/2

Paper 2

Monday
15/09/2008

1 hour 30 mins
60 marks

Additional material:
 Data Booklet

INSTRUCTIONS TO CANDIDATES

Write your *name*, *CT group* and *candidate number* in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided on this question paper.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

You are advised to show all working in calculations.

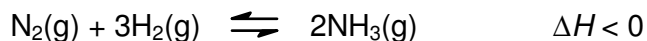
You are reminded of the need for good English and clear presentation in your answers.

For Examiner's Use	1	2	3	4	Total

This question paper consists of **14** printed pages.

[Turn over

- 1 The following data refer to the Haber Process for the manufacture of ammonia.



The table below shows the percentage of ammonia by volume in equilibrium mixtures at various temperatures and pressures. In all cases, N_2 and H_2 were mixed in a 1:3 molar ratio.

Pressure / atm	Temperature		
	300°C	400°C	500°C
1	2.18	0.44	0.13
10	14.7	3.85	1.21
100	51.2	25.1	10.4

- (a) With reference to the data above, explain how changes in temperature and pressure affect the percentage of NH_3 in the equilibrium mixture.

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- (b) Calculate K_p at 100 atm and 400°C.

(c)

- (i) Using the ideal gas equation, sketch two graphs in the axis below at 0°C and 100°C respectively for a fixed mass of gas. Label your graphs clearly.



- (ii) In a 5.0 dm^3 cylinder, 8.0 g of ammonia at 0°C exerts a pressure, p . Calculate the pressure, p .

- (iii) In fact, the pressure inside the cylinder is $1.9 \times 10^5 \text{ Pa}$ under these conditions. Explain why this differs from the pressure you calculated in (ii).

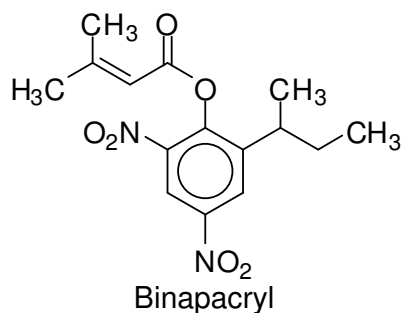
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- (d) Ammonia is used in the Ostwald Process to manufacture nitric acid, which is widely used in the production of fertilizers & some fungicides. One such example is Binapacryl. Its structure is given below.



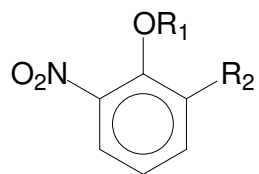
The last step in the manufacture of this fungicide involves the substitution of nitro groups into the benzene ring.

- (i) Draw a dot-and-cross diagram of nitric acid, HNO_3 .

- (ii) State the mechanism for this nitration.

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- (iii) With the aid of curved arrows, describe the mechanism of the nitration in the last step of the manufacture of Binapacryl, starting with



where R_1 : $(\text{CH}_3)_2\text{C}=\text{CHCO}-$ and R_2 : $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)-$.

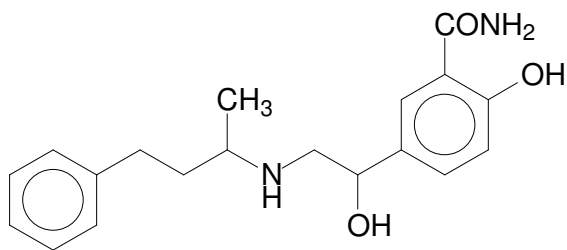
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- (a) Labetalol is a drug used to treat high blood pressure. Its structure is shown below.



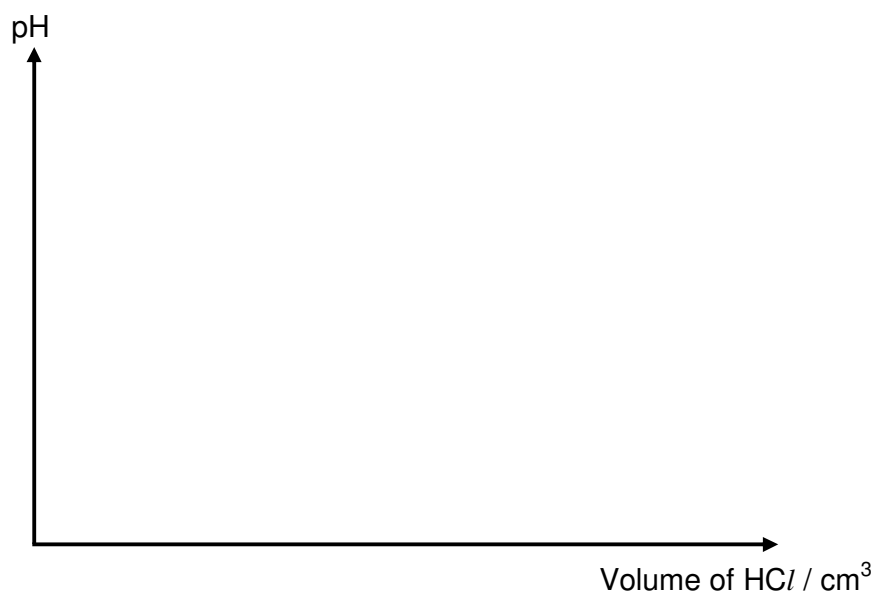
Labetalol

Labetalol is a weak monoacidic alkali with pK_b value 5.30 at 25°C . In one experiment, 25.0 cm^3 of labetalol was titrated against 0.10 mol dm^{-3} of dilute HCl. It was found that 12.60 cm^3 of HCl was required for complete reaction.

- (i) Calculate the concentration of labetalol used.

- (ii) Calculate the pH of the labetalol solution.

- (iii) Sketch a graph to show how the pH changes during the titration. Indicate clearly on your graph the initial pH, the equivalence point and the point of maximum buffering capacity.



- (iv) State a suitable indicator for this titration. Explain your choice.

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- (v) Write equations to show how an aqueous mixture of labetalol (represented as B) and its salt (BH^+Cl^-) can act as a buffer solution

(I) on the addition of a small amount of acid, and

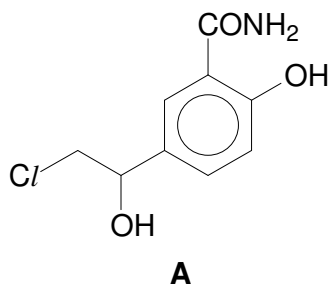
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(II) on the addition of a small amount of alkali.

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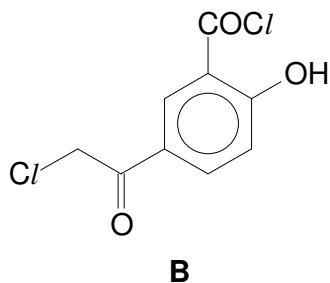
[9]

- (b) Labetalol can be prepared from compound **A** below.



- (i) Explain why an aqueous solution of **A** is acidic. Write an equation to illustrate your answer.

- (ii) Suggest whether compound **B** is more or less acidic than **A** when dissolved in water. Explain briefly with the aid of an equation.



- (iii) Suggest a simple chemical test to distinguish **A** from **B**. Write balanced equation(s) for the reaction that occurs. State what you would observe for each compound.

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[Total: 15]

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- (a) Calcium is a fairly soft, silvery-grey metal whereas iron is a hard, silvery-white metal. Below are some data on calcium and iron.

metal	electronic configuration	relative atomic mass	atomic radius / nm	density / g cm ⁻³
Ca	[Ar]4s ²	40	0.197	1.54
Fe	[Ar]3d ⁶ 4s ²	56	0.116	7.86

Account for the larger density of iron compared to calcium.

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- (b) Calcium and magnesium are both Group II elements. They form nitrates which can be thermally decomposed.

(i) Write the equation for the thermal decomposition of $\text{Mg}(\text{NO}_3)_2$.

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(ii) Which compound, $\text{Ca}(\text{NO}_3)_2$ or $\text{Mg}(\text{NO}_3)_2$, has a higher decomposition temperature? Explain your answer.

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(iii) Given that the decomposition of $\text{Mg}(\text{NO}_3)_2$ is an endothermic process, predict and explain the sign of ΔG for the reaction at high temperatures.

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(c) The table below shows the pH values of some Period 3 chlorides in water.

Compound	pH of a 1.0 mol dm^{-3} solution in water
NaCl	7.0
MgCl_2	6.0
AlCl_3	3.0

(i) With the aid of an equation, explain why an aqueous solution of MgCl_2 is slightly acidic.

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- (ii) Explain why the pH of an aqueous solution of $AlCl_3$ is much lower than that of $MgCl_2$.

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- (iii) Using the data given in the table in (c) together with further data from the *Data Booklet*, suggest the pH of 1.0 mol dm^{-3} solution of $CaCl_2$. Explain your answer.

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- (iv) Describe the observations when sodium hydroxide is added until in excess to an aqueous solution of aluminium chloride. Write balanced equations for the reactions that occur.

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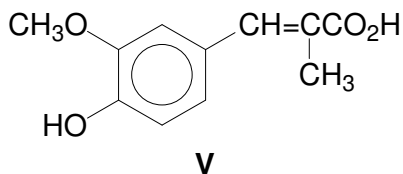
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- 4 Compound **V**, with molecular formula $C_{11}H_{12}O_4$, has the structure shown below.



[The CH_3O- group can be considered to be relatively inert.]

(a)

- (i) Name the isomerism exhibited by **V** and draw the isomers.

- (ii) When gaseous hydrogen chloride is reacted with **V**, the product exhibits a different type of isomerism. Name the isomerism and draw the isomers.

- (iii) Compound **W**, a structural isomer of **V**, gives a yellow precipitate on warming with alkaline aqueous iodine. Suggest a possible structural formula for **W**.

(b) Give the structural formulae of the organic products formed when **V** is added to

(i) $\text{Br}_2(\text{aq})$,

(ii) CH_3COCl , and

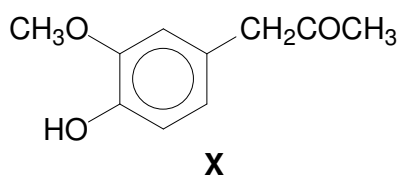
(iii) H_2 with Pt catalyst.

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(c) Suggest why the solubility of **V** in water increases on addition of aqueous NaOH.

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- (d) Propose a three-step synthetic route for the formation of **V** from **X**. In your answer, suggest the reagents and conditions involved in each step and the structures of the intermediates formed.



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[Total: 15]